

ServiceNow Project Documentation

Project: Laptop Request & Approval Application

1. Introduction

This project focuses on developing a custom ServiceNow application to manage laptop requests and manager approvals. The application uses a custom table, Record Producer, Catalog Item, and Approval Relationship to create an end-to-end request process.

The major components developed are:

- Custom Update Set
- Custom Application
- Laptop Request Form Table
- Approval Relationship
- Record Producer
- Employee Service Catalog
- Catalog Variables and Field Mapping
- Manager Approval Integration

2. Creation of a New Update Set

An Update Set is used to capture configuration changes made during application development so that they can be moved between ServiceNow instances.

Steps

Navigate to All → System Update Sets → Local Update Sets.

Click New.

Enter the following details:

Field	Value
Name	HR Onboarding Application
State	In Progress
Release Date	Required release date
Short Description	HR onboarding application configuration

Click Submit.

Open the newly created Update Set.

Click Make Current.

Purpose

The Update Set captures the configuration changes related to the application and helps maintain the changes as a single development package.

3. Creation of Custom Application

A custom application was created using ServiceNow Studio.

Steps

Navigate to All → ServiceNow Studio.

Click Create Application.

Enter the application name:

Project Application

Select the application scope as:

Scoped

Click Continue.

Purpose

The custom application provides a dedicated scope for developing the Laptop Request functionality and its related components.

4. Creation of Laptop Request Form Table

A custom table was created inside the Project Application to store laptop request information.

Steps

Open the Project Application in ServiceNow Studio.

Navigate to Data → Tables.

Click New.

Create the required table.

Define the required fields according to the application requirements.

Configure the field types appropriately.

Save the table.

Table

Laptop Request Form

Example table name:

u_laptop_request_form

Purpose

The Laptop Request Form table stores the information submitted by employees when requesting a laptop.

5. Approval Relationship

Purpose of Approval Relationship

The approval relationship is used to connect the Laptop Request Form table with ServiceNow's approval table:

sysapproval_approver

This allows approval records associated with a laptop request to be displayed along with the request.

The relationship enables the application to support a manager approval process for laptop requests.

6. Creating the Relationship

Steps

Navigate to All → Relationships.

Click New.

Enter the relationship details.

Field	Value
Name	Laptop Request
Applies to table	Laptop Request Form

Field	Value
Queries from table Approval	(sysapproval_approver)

Add the required relationship script.

Click Submit.

Relationship Logic

The relationship script is configured so that approval records associated with the laptop request can be retrieved and displayed.

7. Configuring the Relationship on the Table

After creating the relationship, it needs to be added to the Laptop Request Form table.

Steps

- Open the Laptop Request Form table.
- Navigate to the table configuration.
- Open Related Lists / Relationships configuration.
- Locate the created Laptop Request relationship under the available relationships.
- Add it to the selected relationships.
- Save the configuration.

Result

The Approval records are now available as a related list on the Laptop Request Form.

This allows users to view the approval information associated with a laptop request.

8. Creating a Record Producer

A Record Producer is used to provide a user-friendly form for creating records in a ServiceNow table.

Navigation

Service Catalog → Catalog Definitions → Record Producers

Steps

Navigate to Service Catalog → Catalog Definitions → Record Producers.

Click New.

Enter the following details:

Field	Value
Name	Laptop Request
Table	Laptop Request Form
Table Name	u_laptop_request_form

Add the required variables.

Configure the variable/question types according to the corresponding table fields.

Map the variables to the appropriate fields.

Save the Record Producer.

9. Adding Variables to the Record Producer

Variables were added to collect information from the employee.

The variables correspond to the fields available in the Laptop Request Form table.

Example

Variable	Question Type	Mapped Table Field
Employee Name	Single Line Text	Employee Name
Department	Choice	Department
Laptop Type	Choice	Laptop Type
Justification	Multi Line Text	Justification
Manager	Reference	Manager

The actual variables can be modified according to the requirements of the application.

Purpose

The variables allow employees to enter their request information through a simple Service Catalog form.

The submitted values are mapped to the corresponding fields in the Laptop Request Form table.

10. Creating Employee Service Catalog Item

A Catalog Item was created to organize and expose the Laptop Request functionality through the Service Catalog.

Steps

Navigate to Service Catalog → Catalog Definitions → Maintain Items.

Click New.

Create the catalog item with the following details:

Field	Value
Name	Employee Service Catalog
Category	Service

Save the Catalog Item.

11. Adding Record Producer to Employee Service Catalog

The Laptop Request Record Producer was added to the Employee Service Catalog.

Steps

Open the Employee Service Catalog.

Configure the available catalog content.

Add the Laptop Request Record Producer.

Save the configuration.

Result

Employees can access the Laptop Request functionality from the Employee Service Catalog.

12. Application Workflow

The overall application flow is:

```
Employee Service Catalog
↓
Laptop Request Record Producer
↓
Laptop Request Form
```

↓
Laptop Request Record Created
↓
Approval Relationship
↓
Manager Approval
↓
Approval Status Updated

13. Key ServiceNow Components Used

Component	Purpose
Update Set	Captures application configuration changes
ServiceNow Studio	Used to create and manage the custom application
Scoped Application	Provides an isolated application scope
Custom Table	Stores laptop request information
Relationship	Connects laptop requests with approval records
sysapproval_approver	Stores approval information
Record Producer	Provides a user-friendly request form
Variables	Collect request information from users
Catalog Item	Organizes the service in the Service Catalog
Service Catalog	Provides access to employee services

14. Expected Outcome

After completing the configuration, an employee can access the Employee Service Catalog and submit a Laptop Request.

The submitted information is stored in the Laptop Request Form table. The configured relationship connects the request with the ServiceNow approval records, allowing the request to go through the required manager approval process.

This provides a structured approach for managing laptop requests within ServiceNow.

15. Conclusion

The Laptop Request & Approval Application demonstrates the implementation of a custom ServiceNow application using multiple platform features. The project integrates a custom table, Record Producer, Service Catalog, application scope, table relationships, and approval functionality.

The implementation provides a foundation for automating the employee laptop-request process and can be further enhanced by adding Flow Designer workflows, notifications, approval rules, UI policies, data validation, and automated status updates.